ARJUN MEHTA

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PROFESSIONAL SUMMARY

I am a computer science graduate focused on backend engineering, distributed systems, and infrastructure that remains predictable under failure. I have built high-throughput services in Go and Java, implemented event-driven workflows with Kafka, and improved production reliability through tracing, load testing, and safer deployment automation. My strongest work combines systems fundamentals with practical delivery: clear APIs, measurable performance, and operational documentation. I am looking for backend or platform engineering roles where I can contribute to scalable services and continue developing expertise in distributed architecture.

EDUCATION

Indian Institute of Technology Hyderabad - Bachelor of Technology - BTech, Computer Science and Engineering — May 2026 | Bengaluru, Karnataka

Relevant Coursework: Algorithms, Operating Systems, Computer Networks, Distributed Systems, Databases, Compilers, Cloud Computing, Applied Cryptography

EXPERIENCE

Backend Engineering Intern | NimbusCart Technologies — January 2026 - June 2026 | Bengaluru, India

- Developed Go services for order-state transitions and inventory reservations, handling 1.8 million daily events through Kafka consumers with retry and dead-letter workflows.
- Reduced duplicate reservation incidents by 76% using idempotency keys, transactional outbox patterns, and reconciliation jobs.
- Improved 99th-percentile checkout latency from 1.6 seconds to 780 ms by profiling database access, adding composite indexes, and parallelizing independent calls.
- Created load tests and OpenTelemetry traces used in release gates for three critical services.

Open-Source Engineering Fellow | CodeCommons Fellowship — May 2025 - November 2025 | Remote

- Contributed streaming backpressure controls, structured error handling, and 140 unit tests to a community Java workflow engine.
- Wrote migration documentation for a breaking configuration update and collaborated with maintainers across four time zones.
- Resolved a race condition in worker shutdown that previously caused intermittent test failures in continuous integration.

Teaching Assistant - Operating Systems | IIT Hyderabad CSE Department — August 2025 - December 2025 | Hyderabad, India

- Supported 96 students in labs covering processes, scheduling, virtual memory, file systems, synchronization, and network sockets.
- Designed debugging exercises for deadlocks and memory corruption and held weekly code-review sessions.

SELECTED PROJECTS

Replicated Commit Log / *Go, gRPC, Raft, BadgerDB, Docker* — August 2025 - December 2025

- Implemented a three-node replicated log with leader election, append entries, snapshots, and recovery from persisted state.
- Built fault-injection tests for partitions, delayed messages, process restarts, and stale leaders across 160 deterministic scenarios.
- Measured sustained throughput of 31,000 append operations per second with quorum acknowledgments on local containers.

Payment Reconciliation Pipeline / *Java, Spring Boot, Kafka, PostgreSQL, Redis* — March 2025 - July 2025

- Created an event-driven pipeline that matched gateway settlements to internal transactions and generated exception queues for missing or conflicting records.
- Added exactly-once business semantics through deduplication tables and compensating updates rather than relying on broker guarantees alone.

ADDITIONAL EXPERIENCE HIGHLIGHTS

- Worked across NimbusCart Technologies, CodeCommons Fellowship, and IIT Hyderabad CSE Department in roles requiring independent implementation, structured review, and clear status communication.
- Combined Go, Java, Apache Kafka with database, cloud, testing, and version-control workflows to move projects from initial requirements through documented validation.
- Balanced performance and reliability goals by measuring latency, throughput, data quality, or user completion before and after technical changes.
- Prepared design notes, test plans, demos, and handoff documentation so classmates, researchers, and internship teammates could reproduce results and maintain the work.

PROJECTS

Semantic Code Search / *Python, FastAPI, FAISS, Tree-sitter, React* January 2025 - April 2025

- Indexed functions from 420 public repositories using syntax-aware chunking and hybrid lexical/vector retrieval.
- Improved top-five relevance by 19 percentage points over a keyword baseline on a manually labeled query set.

Campus Shuttle Tracker / *Go, WebSockets, TimescaleDB, Flutter* September 2024 - December 2024

- Processed simulated GPS updates, estimated arrival times, and streamed route changes to mobile clients with reconnect support.
- Designed retention policies and downsampling queries for historical trip analysis.

SELECTED TECHNICAL CONTRIBUTIONS

- Designed project architecture for Replicated Commit Log and Payment Reconciliation Pipeline, separating interface, application, data, and infrastructure concerns to support iterative changes.
- Implemented validation and error-handling paths for expected inputs, malformed records, partial failures, retries, and recovery scenarios rather than testing only the happy path.
- Used automated unit, integration, or hardware/system tests to verify core behavior and added repeatable fixtures or seeded datasets for regression testing.
- Profiled or benchmarked critical workflows and used measured results to prioritize indexing, caching, batching, concurrency, memory, or communication improvements.
- Created deployment or build automation with version control and documented environment setup, configuration, observability, and rollback expectations.
- Presented technical tradeoffs to peers and nontechnical stakeholders, connected implementation choices to user or operational outcomes, and incorporated review feedback.

LEADERSHIP & ACTIVITIES

- Systems Club Coordinator - organized talks on Linux internals, networking, distributed consensus, and performance profiling.
- Inter-IIT Tech Meet participant - contributed backend services and deployment automation for a campus mobility challenge.
- Peer mentor - reviewed resumes and mock technical interviews for second- and third-year students.

TECHNICAL SKILLS

Languages: Go, Java, Python, C++, SQL, Bash

Backend: gRPC, Spring Boot, REST, Kafka, Redis, PostgreSQL, TimescaleDB

Systems: Concurrency, distributed systems, Linux, networking, profiling, fault injection

Cloud & DevOps: Docker, Kubernetes, AWS, GitHub Actions, Terraform, OpenTelemetry, Prometheus

Testing: JUnit, Go testing, integration tests, load tests, chaos simulations

ACADEMIC APPLICATIONS

- Applied Algorithms, Operating Systems, and Computer Networks concepts when selecting data structures, algorithms, interfaces, and performance constraints.
- Used ideas from Distributed Systems and Databases to design storage models, data flows, communication boundaries, or experiment plans.
- Connected classroom work in Compilers to maintainable code organization, testing strategy, documentation, and team review practices.
- Completed technical writing and presentation deliverables explaining assumptions, limitations, evaluation methods, and next-step recommendations.
- Maintained reproducible repositories with setup instructions, sample data, issue tracking, and versioned changes suitable for independent review.

CERTIFICATIONS

- AWS Certified Developer - Associate, 2026
- Linux Foundation: Introduction to Kubernetes, 2025

HONORS & AWARDS

- Institute Academic Excellence Scholarship, 2024-2026
- Inter-IIT Tech Meet Finalist, 2025

ADDITIONAL DETAILS

Professional Interests: production-quality systems, measurable performance, careful testing, accessible documentation, and cross-functional delivery.

Profile Status: Synthetic testing profile; no real employment, credential, or identity claim is represented.

Links: github.com/arjunmehta-sample | www.linkedin.com/in/arjun-mehta-backend-sample | arjunmehta.example.com